Spline Baseline

This document illustrates the baseline fitting with the spline fit. For that make_lsq_spline is used.

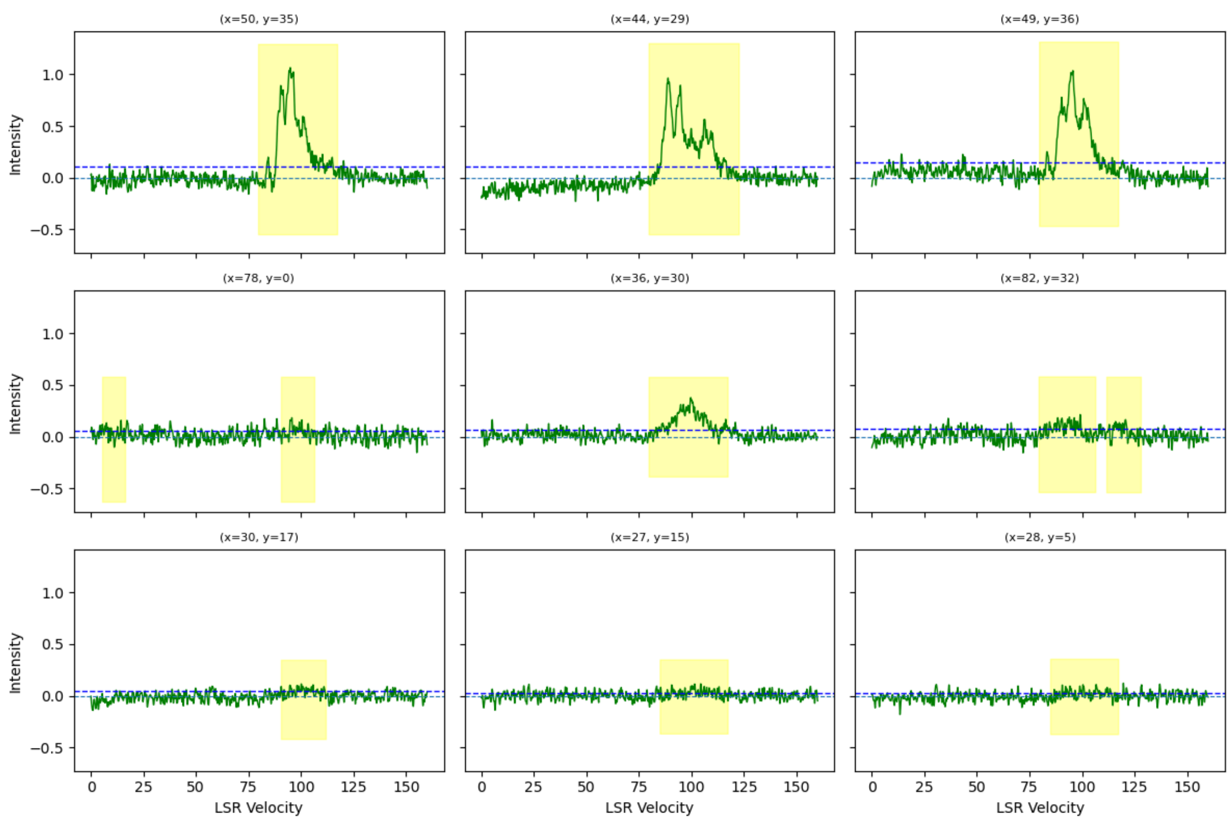
A spline is a piecewise polynomial function whose flexibility is controlled by knot locations.

For the baseline fitting using the **spline** HCN cube is used in this document. HCN cube is spectrally trimmed to range [0,160] km/s.

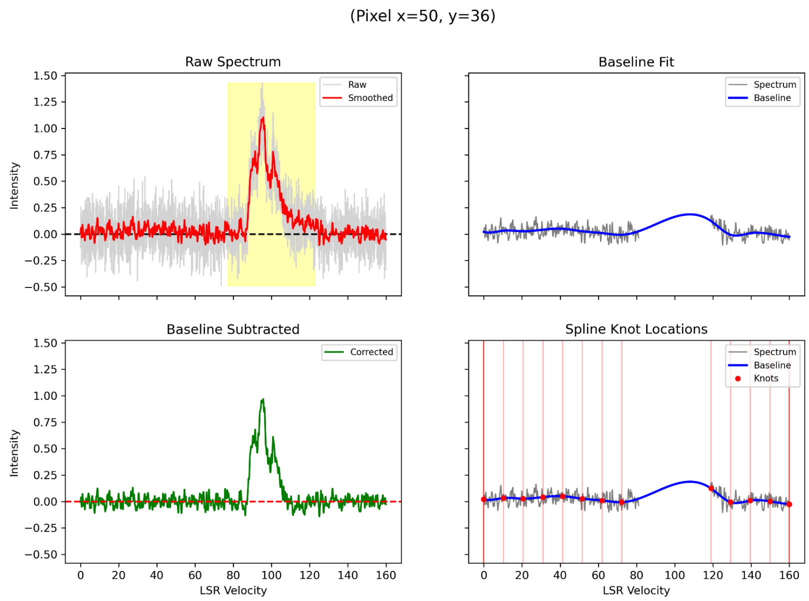
The masking is done using the automatic masking method applied to binned spectra.

This plot represents the spectra at 3 **strong** pixels (top row), 3 **moderate** pixels (middle row), and 3 **weak** pixels (bottom row).

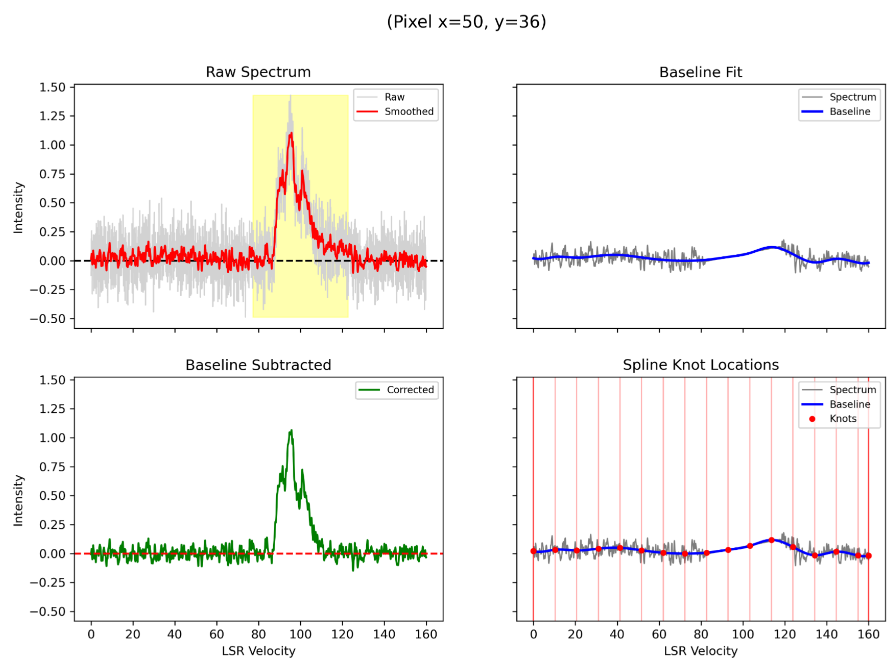
The blue dashed line represents the threshold value for the masking condition. (threshold is calculated as, threshold = mean + std*clip_value.
The green dashed line is 0 line
The yellow shaded region represents the emission region.



The emission channels identified by the mask are excluded before knot placement. Interior knots are then placed only within the remaining baseline channels at intervals defined by knot_spacing. For broad emission features, this results in large gaps between neighbouring knots, reducing the local constraints on the spline.

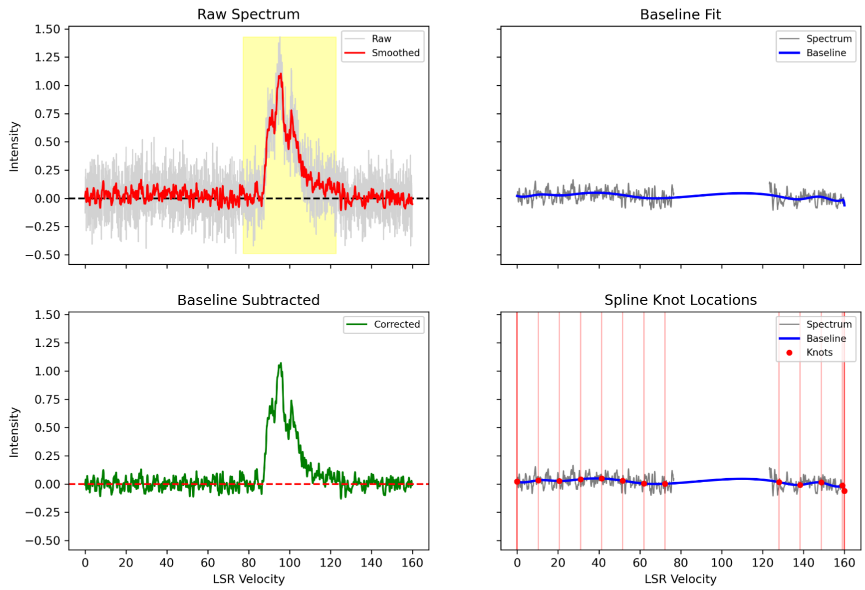


To reduce this effect, the masked emission region can be replaced with a straight line whose slope is estimated from the neighbouring baseline channels.



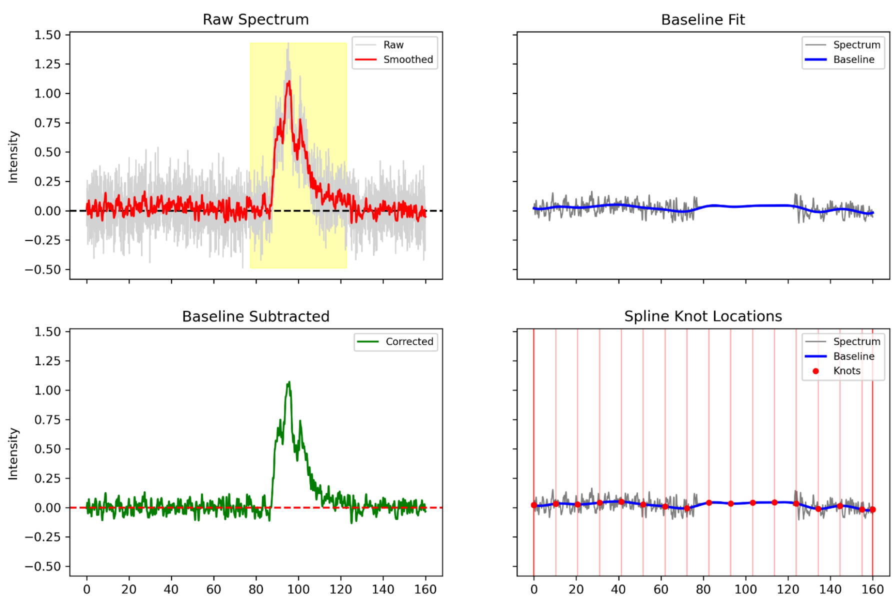
Another possibility for the bump can be that the emission wings not being included in the emission mask can produce a curvature in the baseline. To deal with this we can expand the emission to a few channels or bins, although it is possible that it might not be enough for more broader wings in some pixels.

(Pixel x=50, y=36)



A more conservative approach is to combine both interpolation and mask dilation.

(Pixel x=50, y=36)



Summary

- Broad emission regions may produce artificial bumps or dips because no knots are available inside the masked region.
- Replacing the masked region with an interpolated straight line allows knots to be retained and improves spline stability.
- Expanding the mask helps protect against emission wings contaminating the baseline fit.
- Combining interpolation and mask dilation provides the most conservative treatment.

In the above plots parameters used are

knot_start : 200
knot_spacing : 200
Bin_expand : 2 (in case of dilation)
Spline order : 3 (default)

Note: These parameters are in case of HCN (w43) for different regions and line parameters can be modified.